As a specific example, consider a spring mass system (or oscillation of pendulum) as shown in Figure 10.8. When the spring moves up and down (or pendulum moves to and fro), the motion of the mass or bob is mapped to points on the circular motion.

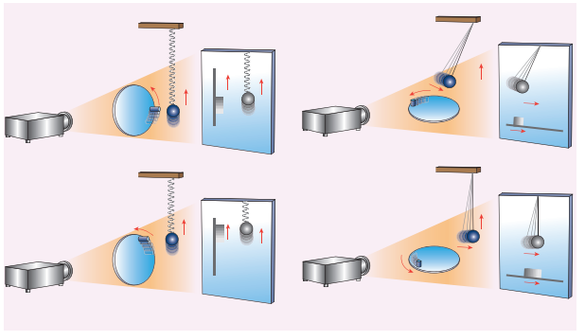


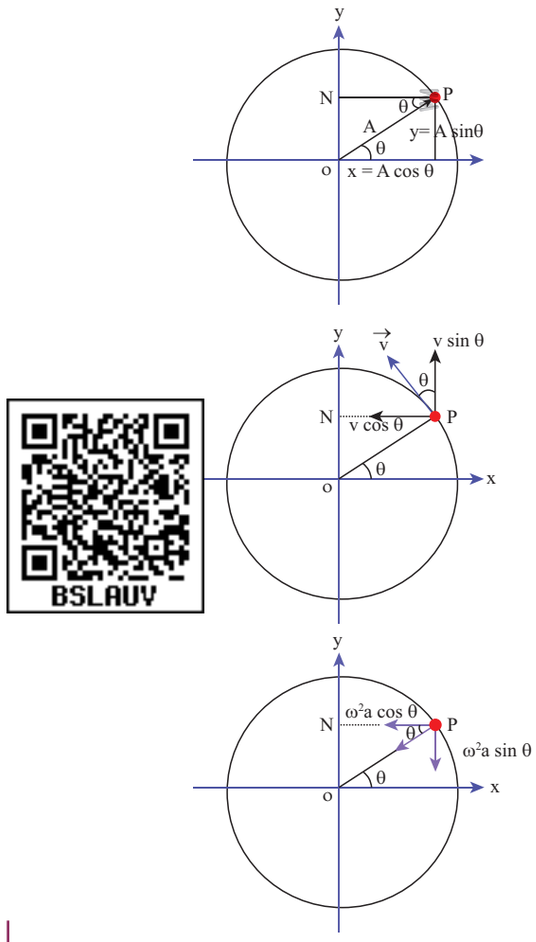
Figure 10.8 Motion of spring mass (or simple pendulum) related to uniform circular motion

Thus, if a particle undergoes uniform circular motion then the projection of the particle on the diameter of the circle (or on a line parallel to the diameter) traces straightline motion which is simple harmonic in nature. The circle is known as reference circle of the simple harmonic motion. *The simple harmonic motion can also be defined as the motion of the projection of a particle on any diameter of a circle of reference.*

Activity

a. Sketch the projection of ***spiral in*** motion as a wave form.
b. Sketch the projection of ***spiral out*** motion as a wave form.

10.2.2 Displacement, velocity, acceleration and its graphical representation - SHM



BSLAUV

Figure 10.9 Displacement, velocity and acceleration of a particle at some instant of time

The distance travelled by the vibrating particle at any instant of time t from its mean position is known as displacement.

Let *P* be the position of the particle on a circle of radius A at some instant of time *t* as shown in Figure 10.9. Then its displacement y at that instant of time *t* can be derived as follows In ΔOPN

$$\sin\theta = \frac{ON}{OP} \Rightarrow ON = OP\sin\theta \qquad (10.5)$$

But $\theta = \omega t$, ON = y and OP = A

UNIT-10(XI-Physics_Vol-2).indd 193 17-08-2020 16:23:24